

On a smart contract platform, the possibilities rapidly expand beyond what developers desiring to integrate various applications can easily handle. This leads to the adoption of standard interfaces for different types of functionality. On Ethereum, these standards are called *Ethereum Request for Comments* (ERC). The best known of these define different types of tokens that have similar behavior. ERC-20 is the standard for fungible tokens and defines an interface for tokens whose units are identical in utility and functionality.² It includes behavior such as transferring units and approving operators for using a certain portion of a user's balance. Another is ERC-721, the non-fungible token standard, which are unique and often used for collectibles or assets such as peer-to-peer loans. The benefit of these standards is that application developers can code for one interface and support every possible token that implements that interface. We will discuss these interfaces in more detail later on.

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An interesting problem with blockchain protocols is that they are isolated from the world outside of their ledger. That is, the Ethereum blockchain authoritatively knows what is happening only on the Ethereum blockchain and not, for example, the level of the S&P 500 or which team won the Super Bowl. This limitation constrains applications to Ethereum native contracts and tokens, thus reducing the utility of the smart contract platform; it is generally known